

Math 534H - Spring 2026 - Lecture 24

①

zero Dirichlet BC. for wave equation on $(0, l)$

Using ① on page 9 Lecture 23 to find the

coefficients A_n , $n \geq 1$ such that

$$\phi(x) = \sum_{n=1}^{\infty} A_n \cdot \sin\left(\frac{n\pi x}{l}\right)$$

Multiply both sides by $\sin\left(\frac{m\pi x}{l}\right)$ and integrate

both sides between 0 and l . Assuming

we can justify (which we can!) $\int_0^l \sum_{n=1}^{\infty} \dots = \sum_{n=1}^{\infty} \int_0^l \dots$

We get from (1) that for any $m \geq 1$

$$A_m := \frac{2}{l} \int_0^l \phi(x) \cdot \sin\left(\frac{m\pi x}{l}\right) dx.$$

Remark: as long as $\phi(x)$ is "nice" so that

$\phi(x) \cdot \sin\left(\frac{m\pi x}{l}\right)$ is integrable then the

A_m are well defined constants (could be 0)
for $m \geq 1$. (of course)

Note: If we are considering the heat/diffusion equation with zero Dirichlet conditions, then.



is also how you find the coefficients of the sine series for the initial temperature or initial concentration $\phi(x)$.

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Back to the wave: how do we find the coefficients

B_n such that

$$\psi(x) = \sum_{n=1}^{\infty} B_n \frac{n\pi c}{l} \sin\left(\frac{n\pi x}{l}\right)$$

? ↗ wave speed.

Using ① and repeating the steps we did

above we get

(4)

$$B_m \cdot \frac{m\pi c}{l} = \frac{2}{l} \int_0^l \psi(x) \sin\left(\frac{m\pi x}{l}\right) dx$$

So that ($m \neq 0$)

$$B_m := \frac{2}{m\pi c} \int_0^l \psi(x) \sin\left(\frac{m\pi x}{l}\right) dx$$

Next: recall that when solving the wave or the heat/diffusion equations with zero Neumann BC we were led to the question of whether

the initial position $\phi(x)$ (wave) or initial temperature $\phi(x)$ or concentration $\phi(x)$ (heat/diffusion); and the initial velocity $\psi(x)$ (wave) could be represented as cosine series $x \in (0, l)$

$$\phi(x) \stackrel{?}{=} \frac{1}{2} A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{l}\right) \quad (+)$$

$$\psi(x) \stackrel{?}{=} \frac{1}{2} B_0 + \sum_{n=1}^{\infty} B_n \frac{n\pi c}{l} \cos\left(\frac{n\pi x}{l}\right) \quad (++)$$

↗ wave speed

Now using (2) page 9 Lecture 23 and proceeding as above and integrating both sides of (+) against

the $\cos\left(\frac{m\pi x}{l}\right)$ for $m \geq 0$ we get: For $m=0$ ⁽⁶⁾

$$A_0 = \frac{2}{l} \int_0^l \phi(x) \cdot \underbrace{\cos\left(\frac{0\pi x}{l}\right)}_{=1} dx \Rightarrow$$

$$A_0 = \frac{2}{l} \int_0^l \phi(x) dx$$

For $m \geq 1$ we get

$$A_m = \frac{2}{l} \int_0^l \phi(x) \cdot \cos\left(\frac{m\pi x}{l}\right) dx$$

(7)

Next integrating Ψ against the $\cos\left(\frac{n\pi x}{l}\right)$ $n \geq 0$
 we find the coefficients B_0 and B_n for the
 velocity Ψ . We get:

$$B_0 = \frac{2}{l} \int_0^l \Psi(x) \cdot dx \quad \left(\text{using } \cos\left(\frac{0 \cdot \pi x}{l}\right) = 1 \right)$$

$$\frac{n\pi c}{l} B_n = \frac{2}{l} \int_0^l \Psi(x) \cdot \cos\left(\frac{n\pi x}{l}\right) dx, \text{ so}$$

$$B_n = \frac{2}{n\pi c} \int_0^l \Psi(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad \underline{n \geq 1}$$

Some examples (Section 5.1 of Strauss):

(8)

① Let $\phi(x) = 1$ for all $x \in (0, l)$. Find:

① Cosine Series

$$A_0 = \frac{2}{l} \int_0^l \phi(x) dx = \frac{2}{l} \int_0^l 1 dx = 2$$

For $m \geq 1$

$$A_m = \frac{2}{l} \int_0^l \phi(x) \cos\left(\frac{m\pi x}{l}\right) dx$$

$$= \frac{2}{l} \int_0^l \cos\left(\frac{m\pi x}{l}\right) dx, \text{ but the}$$

cosine is an even function. So

$$\int_{-l}^l \cos(z) dz = 2 \int_0^l \cos(z) dz. \quad (*)$$

Now by (6) page 10 of Lecture 23 we have that

$$\int_{-l}^l \cos\left(\frac{m\pi x}{l}\right) dx = 0 \Rightarrow \int_0^l \cos\left(\frac{m\pi x}{l}\right) dx = 0 \quad (*)$$

So: $A_m = \frac{2}{l} \int_0^l \cos\left(\frac{m\pi x}{l}\right) dx = 0$ for all $m \geq 1$

Then

$$1 = \frac{\frac{A_0}{2}}{2} + 0 \cos\left(\frac{1\pi x}{l}\right) + 0 \cos\left(\frac{2\pi x}{l}\right) + 0 \cos\left(\frac{3\pi x}{l}\right) + \dots$$

ii

Sine series : now we only have $m \geq 1$ and (10)

$$A_m = \frac{2}{l} \int_0^l \underbrace{\phi(x)}_{=1} \cdot \sin\left(\frac{m\pi x}{l}\right) dx \Rightarrow$$

$$A_m = \frac{2}{l} \int_0^l \sin\left(\frac{m\pi x}{l}\right) dx = \frac{2}{l} \frac{l}{m\pi} \left(-\cos\left(\frac{m\pi x}{l}\right) \right) \Big|_{x=0}^{x=l}$$

$$= -\frac{2}{m\pi} \left[\cos(m\pi) - \cos(0) \right]$$

$$= -\frac{2}{m\pi} \left[(-1)^m - 1 \right]$$

Ans

$$A_m = \begin{cases} 0 & \text{if } m \text{ is EVEN} \\ \frac{4}{m\pi} & \text{if } m \text{ is ODD} \end{cases} \Rightarrow$$

$$1 = \frac{4}{\pi} \sin\left(\frac{\pi x}{l}\right) + \frac{4}{3\pi} \sin\left(\frac{3\pi x}{l}\right) + \frac{4}{5\pi} \sin\left(\frac{5\pi x}{l}\right) + \dots$$

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$$1 = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{(2k-1)} \sin\left(\frac{(2k-1)\pi x}{l}\right)$$

2 Consider $\phi(x) = x$ and $l=1$, so $x \in (0, 1)$

i Cosine Series

ii Sine Series

i $n \geq 0$, so first $n=0$ gives

$$A_0 = 2 \int_0^1 x \, dx = 2 \cdot \frac{1}{2} x^2 \bigg|_{x=0}^{x=1} = 1$$

For $m \geq 1$ $A_m = 2 \int_0^1 x \cos(m\pi x) dx \Rightarrow$ (12)

$$A_m = \left[\frac{2x}{m\pi} \sin(m\pi x) + \frac{2}{m^2\pi^2} \cos(m\pi x) \right] \Bigg|_{x=0}^{x=1}$$

IBP

$$A_m = \frac{2}{m^2\pi^2} \left[(-1)^m - 1 \right] = \begin{cases} 0 & \text{if } m \text{ is even} \\ -\frac{4}{m^2\pi^2} & \text{if } m \text{ is odd} \end{cases}$$

Thus on the interval $(0, 1)$

$$x = \frac{1}{2} - \frac{4}{\pi^2} \cos(\pi x) - \frac{4}{9\pi^2} \cos(3\pi x) - \frac{4}{25\pi^2} \cos(5\pi x) - \dots$$

(infinite series)

That is

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$$x = \frac{1}{2} - \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} \cos((2k-1)\pi x).$$

ii) For the sine series we only have $n \geq 1$ so

$$A_n = 2 \int_0^1 x \cdot \sin(n\pi x) dx \underset{\text{IBP}}{=} (-1)^{n+1} \cdot \frac{2}{n\pi}$$

Hence: on the interval $(0, 1)$ we have that

$$x = \frac{2}{\pi} \sin(\pi x) - \frac{1}{\pi} \sin(2\pi x) + \frac{2}{3\pi} \sin(3\pi x) - \frac{1}{2\pi} \sin(4\pi x) + \dots$$

(infinite series)

Factoring out $\frac{2}{\pi}$ we can rewrite as :

(14)

$$X = \frac{2}{\pi} \cdot \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \sin(m\pi x)$$

3

Solve the problem :

$$\left\{ \begin{array}{l} u_{tt} - u_{xx} = 0 \quad : \text{Wave Equation} \\ u(0,t) = 0 = u(3,t) \quad : \text{zero Dirichlet} \\ u(x,0) = x, \quad u_t(x,0) = 0 \end{array} \right\} *$$

In this problem $c=1$, $l=3$

We know that for zero Dirichlet BC the solution of

(*) has the form:

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos\left(\frac{n\pi t}{3}\right) + B_n \sin\left(\frac{n\pi t}{3}\right) \right) \cdot \sin\left(\frac{n\pi x}{3}\right)$$

We now need to find A_n, B_n so that $u(x,0) = x$

and $u_t(x,0) = 0$. Let's start from the latter.

First we compute $u_t(x,t)$:

$$u_t(x,t) = \sum_{n=1}^{\infty} \frac{n\pi}{3} \left(-A_n \sin\left(\frac{n\pi t}{3}\right) + B_n \cos\left(\frac{n\pi t}{3}\right) \right) \sin\left(\frac{n\pi x}{3}\right)$$

As:

$$u_t(x,0) = \sum_{n=1}^{\infty} \frac{n\pi}{3} \cdot B_n \cdot \sin\left(\frac{n\pi x}{3}\right)$$

Since we want $u_t(x, 0) = 0 \Rightarrow$

$$0 = \sum_{n=1}^{\infty} \frac{n\pi}{\ell} B_n \cdot \sin\left(\frac{n\pi x}{3}\right)$$

Multiplying both sides by $\sin\left(\frac{m\pi x}{3}\right)$ for any $m \geq 1$

and integrating in x between 0 and 3, we get that $B_m = 0$ for all $m \geq 1$. Coming back to

$u(x, t)$ we then have (all $B_n = 0$, for all $n \geq 1$)

that at $t = 0$

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{3}\right) = X \underline{\text{so}}$$

Repeating the calculations that we did in example

(2) (ii) in page 14 but on the interval $(0, 3)$ instead

of on the interval $(0, 1)$ we obtain that

(see also page 109 in Strauss book example 3)

$$A_m = \frac{2}{3} \int_0^3 x \sin\left(\frac{m\pi x}{3}\right) dx$$

$$= (-1)^{m+1} \frac{6}{m\pi} \quad \text{so that}$$

$$u(x, 0) = \frac{6}{\pi} \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \sin\left(\frac{m\pi x}{3}\right) = x \quad \text{on } (0, 3).$$